

Engineering E. Wolfpack

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EDUCATION

North Carolina State University, Raleigh, NC

May 20XX

Bachelor of Science in Chemical Engineering, Biomolecular concentration

Minor: Biotechnology, CGPA: X.X

RESEARCH EXPERIENCE

North Carolina State University, Raleigh, NC

Jan 20XX-Present

Undergraduate Research Assistant, Wei Research Group

- Designed and developed a nucleic acid detection platform utilizing CRISPR-Cas12a
- Collaborated with Ph.D. mentor on experimental design, execution, and data analysis to advance project goals
- Trained and supervised individual students and groups of 3–4 undergraduates in CRISPR-Cas experiment execution and troubleshooting

Durham Technical Community College, Durham, NC

Sept 20XX-Feb 20XX

Undergraduate Research, Sea-Genes

- Isolated bacteriophage genes by PCR, column purification, checked with gel electrophoresis
- Performed isothermal assembly of DNA fragments to insert genes into bacterial plasmid
- Introduced plasmid DNA into bacterial cells and inoculated a culture from a single bacterial colony
- Performed colony screening via colony PCR

Honors Course Research, Microbial Fuel Cell Efficiency

Sept 20XX-Dec 20XX

- Designed a current producing microorganism catalyzed fuel cell
- Determined and optimized electrode material, substrate, pH, and temperature conditions of fuel cell
- Presented research poster at NC American Chemical Society meeting

North Seattle College, Seattle, WA

Jan 20XX- April 20XX

Undergraduate Research, Exploring Mutations of SARS-CoV-2

- Generated phylogenetic trees in GenBank to compare and analyze genome sequences
- Collaborated with three peer researchers to create a hypothesis and design the research objectives
- Summarized research data comparing and contracting mutations in DNA and environmental factors

WORK EXPERIENCE

XOXOXO Therapeutics, Morrisville, NC

May 20XX-July 20XX

MS&T Intern

- Transfected HEK293 cells and performed qPCR analysis to evaluate productivity outcomes
- Supported bioreactor operations for viral capsid production
- Analyzed experimental data and presented findings at weekly MS&T meetings

Durham Technical Community College, Durham, NC

June 20XX-July 20XX

Mathematics, Chemistry and Physics Tutor at the Center for Academic Excellence

- Taught 4-6 students daily the fundamentals of general chemistry, physics, algebra, and precalculus
- Motivated students to set personal goals and supported them in tracking their progress

Teaching Assistant - General Chemistry

- Led two sessions a week to help students succeed in general chemistry
- Promoted active learning in the lab by leading effective discussions about key chemistry concepts

AWARDS

Goodnight Scholar (Fall 2023-Spring 2026)

A merit-based STEM scholarship program that empowers scholars to lead as transformational figures in their communities and globally. The program fosters advocacy for STEM and STEM education through participation in professional, social, and service events throughout the year.

Stutt Research Scholar (Fall 2024-Spring 2025)

One of 10 ChemE undergraduate students selected as a member of the Stutt's Research Scholars Program to build community within the cohort, improve communication skills, learn how to do research and guide participants in pursuing a research career.

AIChE Annual Student Conference 2024

Second place in undergraduate poster competition in Food, Pharmaceutical, and Biotechnology IX; Applied Supercoiled Plasmids in CRISPR-Dx for Viral Gene Detection.

Emerging Researchers National (ERN) Conference in STEM 2025 Travel Award

National annual conference sponsored by the NSF and AAAS. Travel Award provides competitive funding, recognizes student researchers for excellence and promotes diversity in science.

PUBLICATIONS

1. Mohammad, N., Talton, L., Dalgan, S., Hetzler, Z., **Wolfpack, E.** *Ratiometric nonfluorescent CRISPR assay utilizing Cas12a-induced plasmid supercoil relaxation.* Commun Chem 7, 130 (2024)
2. Bagi, M., Jamalzadegan, S., **Wolfpack, E.**, Wei, Q. *CRISPR-Cas based platforms for RNA detection: fundamentals and applications.* Chem. Commun. 61, 72 (2025).
3. Mohammad, N., **Wolfpack, E.**, Tang, Y., et al. *Attomolar Detection of HIV-1 with Label-Free RCA-rCRISPR on Smartphone.* bioRxiv 2025.06.15.659809;

PRESENTATIONS

1. **Wolfpack E.** *Detection of HIV-1 with RCA-rCRISPR on Smartphone*; 2025 AIChE Annual Meeting (Annual Student Conference: Competitions & Events), Poster Presentation #328, **November 2025.**
2. **Wolfpack E.** *Disease detection with Cas12a protein and supercoiled plasmid reporter*; Emerging Researchers National (ERN) in STEM by AAAS, Oral Presentation #392, **April 2025.**
3. **Wolfpack E.** *Utilizing Supercoiled Plasmids in CRISPR-Dx for Viral Gene Detection*; 2024 AIChE Annual Meeting (Annual Student Conference: Competitions & Events), Poster Presentation #220, **October 2024.**
4. **Wolfpack E.** *CRISPR-Cas12a-based diagnostics using plasmid reporters*; NC State Summer 2024 Undergraduate Research & Creativity Symposium, Poster Presentation #47, **July 2024.**
5. **Wolfpack E.** *Productivity Investigation*; Kriya Therapeutics, Poster Presentation, **July 2024.**
6. **Wolfpack E.** and North D. *Environmental Factors for Microbial Fuel Cell Efficiency*; Durham Tech Honors Program Oral Presentations, **December 2022.**
7. **Wolfpack E.** and North D. *Environmental Factors for Microbial Fuel Cell Efficiency*; 127th North Carolina American Chemical Society meeting, Poster Presentation #40, **November 2022.**

VOLUNTEERING

STEM Coach

Inspired future generations of STEM students by leading engaging activities with elementary and middle school students, including after-school tutoring, hands-on demonstrations, workshops, and science fairs

Peer Mentor

Provided academic and personal guidance to mentees through one-on-one and group meetings helping students navigate university resources, foster engagement, and build a sense of belonging

SATELLITE Camp Committee Member

Organized STEM camp events for underrepresented high school students; coordinated with guest lecturers, designed programming, and secured funding to expand opportunities for participants

Lego Brick Build: An Annual Goodnight Scholars Event

Volunteered for 3 years to help organize and lead STEM activities such as hands-on LEGO projects with scientific themes to spark curiosity and make science fun and accessible for kids from Boys and Girls Clubs